Analysis

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TPOT with graph-light search space versus ensemble

2000 total evals

Version 1

Stats Testing

```
## # A tibble: 26 x 9
##    task_id .y.   group1            group2    n1    n2 statistic       p p.signif
##      <int> <chr> <chr>             <chr>  <int> <int>     <dbl>   <dbl> <chr>
##  1    2073 auroc ensemble_random_~ indiv~    21    21      207  7.21e-4 ***
##  2  146818 auroc ensemble_random_~ indiv~    21    21      202  3.15e-4 ***
##  3  146820 auroc ensemble_random_~ indiv~    21    21      182. 2.28e-2 *
##  4  168350 auroc ensemble_random_~ indiv~    21    21      231  9.54e-7 ****
##  5  168757 auroc ensemble_random_~ indiv~    21    21      231  9.54e-7 ****
##  6  168784 auroc ensemble_random_~ indiv~    21    21      231  9.54e-7 ****
##  7  168911 auroc ensemble_random_~ indiv~    21    21      228  4.77e-6 ****
##  8  190137 auroc ensemble_random_~ indiv~    21    21      207  1.56e-3 **
##  9  190146 auroc ensemble_random_~ indiv~    21    21      210  4.26e-4 ***
## 10  190411 auroc ensemble_random_~ indiv~    21    21      231  9.54e-7 ****
## 11  359955 auroc ensemble_random_~ indiv~    21    21      160. 1.22e-1 ns
## 12  359956 auroc ensemble_random_~ indiv~    20    20      185  1.69e-3 **
## 13  359958 auroc ensemble_random_~ indiv~    21    21      215  1.61e-4 ***
## 14  359959 auroc ensemble_random_~ indiv~    21    21      219  6.68e-5 ****
## 15  359960 auroc ensemble_random_~ indiv~    21    21      231  9.54e-7 ****
## 16  359962 auroc ensemble_random_~ indiv~    21    21      187  1.13e-2 *
## 17  359963 auroc ensemble_random_~ indiv~    21    21      231  9.54e-7 ****
## 18  359964 auroc ensemble_random_~ indiv~    21    21      220  5.25e-5 ****
## 19  359965 auroc ensemble_random_~ indiv~    21    21      186  2.65e-3 **
## 20  359968 auroc ensemble_random_~ indiv~    21    21      189  9.02e-3 **
## 21  359969 auroc ensemble_random_~ indiv~    21    21      231  9.54e-7 ****
## 22  359970 auroc ensemble_random_~ indiv~    21    21      231  9.54e-7 ****
## 23  359971 auroc ensemble_random_~ indiv~    21    21      230  1.91e-6 ****
## 24  359972 auroc ensemble_random_~ indiv~    21    21      231  9.54e-7 ****
## 25  359974 auroc ensemble_random_~ indiv~    21    21      227  6.68e-6 ****
## 26  359975 auroc ensemble_random_~ indiv~    21    21      200. 3.7 e-3 **
```

Medians

```
## # A tibble: 26 x 3
##    task_id individual_tpot ensemble_random_100
##      <int>           <dbl>               <dbl>
##  1    2073           0.873               0.880
##  2  146818           0.919               0.924
##  3  146820           0.995               0.997
##  4  168350           0.955               0.962
##  5  168757           0.782               0.787
##  6  168784           0.948               0.961
##  7  168911           0.858               0.870
##  8  190137           0.907               0.928
##  9  190146           0.946               0.954
## 10  190411           0.916               0.922
## 11  359955           0.746               0.749
## 12  359956           0.927               0.933
## 13  359958           0.937               0.944
## 14  359959           0.742               0.752
## 15  359960           0.998               0.999
## 16  359962           0.834               0.842
## 17  359963           0.993               0.996
## 18  359964           0.993               0.993
## 19  359965           0.999               1.00
```

```
## 20  359968          0.910             0.917
## 21  359969          0.824             0.843
## 22  359970          0.892             0.931
## 23  359971          0.996             0.997
## 24  359972          0.979             0.988
## 25  359974          0.847             0.870
## 26  359975          0.995             0.995
```